Github Link:https://github.com/Gokuldurai2006/predicting-customer-chruns/blob/main/Gokulphase3.ipynb14

Phase-3

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Department: Electronic and communication engineering

Date of Submission: 09-05-2025

PREDICTING CUSTOMER CHURN USING MACHINE LEARNING TO UNCOVER HIDDEN PATTERNS

1. Problem Statement

Customer churn poses a significant challenge for businesses, particularly in highly competitive markets such as telecommunications, banking, and SaaS. Retaining existing customers is often more cost-effective than acquiring new ones. This project aims to develop a machine learning model to predict customer churn using historical behavioral and demographic data. By identifying customers likely to churn, businesses can proactively engage them and reduce attrition. This is a **classification problem**, where the target variable indicates whether a customer has churned or not.

2. Abstract

This project focuses on predicting customer churn using machine learning techniques to uncover hidden patterns in customer behavior and interactions. Churn prediction helps companies identify customers at risk of leaving, enabling timely intervention and improving retention strategies. We use publicly available customer data, preprocess it to handle missing values and encode categorical features, and apply exploratory data analysis to understand key trends. We then train multiple classification models, including Logistic Regression, Random Forest, and XGBoost. The best-performing model is evaluated using accuracy, precision, recall, and F1-score. The model is deployed using Streamlit to allow real-time churn predictions via a web interface.

3. System Requirements

Hardware:

- Minimum 8GB RAM
- i5 or equivalent processor (recommended) **Software:**
- Python 3.8+
- Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, shap, streamlit
- IDE: Jupyter Notebook / Google Colab

4. Objectives

- Develop a robust ML model to predict customer churn.
- Identify key features contributing to churn through pattern discovery.
- Deploy the model via a web application to provide predictions in realtime.
- Help businesses improve retention strategies and reduce customer attrition.

5. Flowchart of Project Workflow



6. Dataset Description

- Source: https://www.kaggle.com/datasets/rashadrmammadov/customerchurn-dataset/code
- Type: Public
- Size: ~7,069 rows and 21 columns
- Structure: Includes customer demographics, account information, and service usage

`df.head()` screenshot

```
try:
    df = pd.read_csv('your_dataset.csv')
    print("Dataset loaded successfully!")
    print("\nFirst 5 rows of the dataset:")
    print(df.head())
except FileNotFoundError:
    print("Error: Dataset file not found. Please ensure 'your_dataset.csv' is in the correct directory.")
    exit()
```

Python

```
Dataset loaded successfully!

First 5 rows of the dataset:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

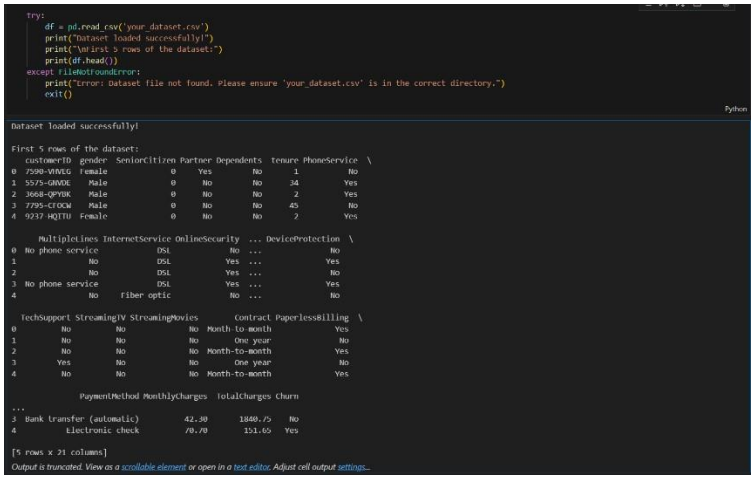
               PaymentMethod MonthlyCharges  TotalCharges Churn
...
3  Bank transfer (automatic)         42.30       1840.75    No
4          Electronic check         70.70        151.65   Yes

[5 rows x 21 columns]
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

7. Data Preprocessing

- Missing values handled using imputation
- Duplicate entries dropped
- Outliers treated using IQR method
- Encoding: One-hot for categorical features
- Feature scaling using StandardScaler

Before Transformation



```
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    print("\nFirst 5 rows of the dataset:")
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except FileNotFoundError:
    print("Error: Dataset file not found. Please ensure 'your_dataset.csv' is in the correct directory.")
    exit()
```

Python

```
Dataset loaded successfully!

First 5 rows of the dataset:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

              PaymentMethod MonthlyCharges  TotalCharges Churn
...
3  Bank transfer (automatic)         42.30       1840.75    No
4       Electronic check             70.70        151.65   Yes

[5 rows x 21 columns]
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

After transformation

```
Processed dataset information after scaling:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   int32
 1   SeniorCitizen     7043 non-null   float64
 2   Partner           7043 non-null   int32
 3   Dependents        7043 non-null   int32
 4   tenure            7043 non-null   float64
 5   PhoneService      7043 non-null   int32
 6   MultipleLines     7043 non-null   int32
 7   InternetService   7043 non-null   int32
 8   OnlineSecurity    7043 non-null   int32
 9   OnlineBackup      7043 non-null   int32
 10  DeviceProtection  7043 non-null   int32
 11  TechSupport       7043 non-null   int32
 12  StreamingTV       7043 non-null   int32
 13  StreamingMovies   7043 non-null   int32
 14  Contract          7043 non-null   int32
 15  PaperlessBilling  7043 non-null   int32
 16  PaymentMethod     7043 non-null   int32
 17  MonthlyCharges    7043 non-null   float64
...
1                 0              3       -0.259629      -0.173244       0
2                 1              3       -0.362660      -0.959674       1
3                 0              0       -0.746535      -0.194766       0
4                 1              2        0.197365      -0.940470       1
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

8. Exploratory Data Analysis (EDA)

- Found that churn is higher among customers with month-to-month contracts and fiber optic internet
- Heatmap showed strong correlation between tenure and churn
- Pie charts and bar plots visualized churn distribution across demographics

Key Insights:

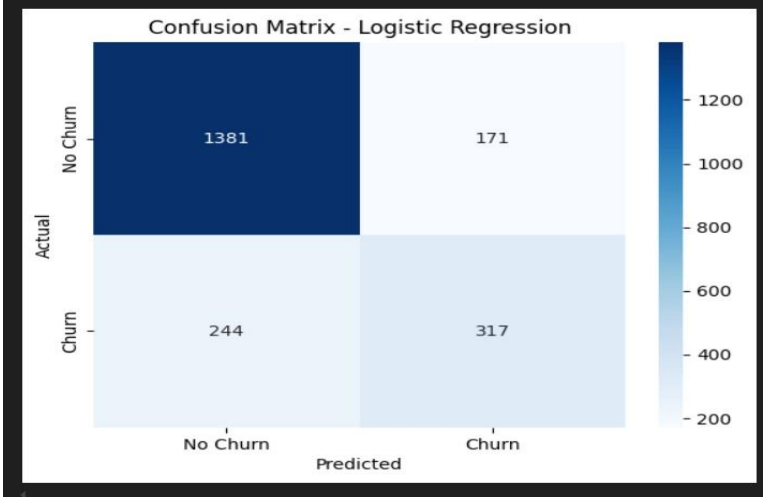
- Shorter **tenure** increases churn risk
- Customers with tech support and longer contracts churn less

```
--- Logistic Regression Evaluation ---
Accuracy: 0.8035967818267865

Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.89      0.87      1552
           1       0.65      0.57      0.60       561

    accuracy                           0.80      2113
   macro avg       0.75      0.73      0.74      2113
weighted avg       0.80      0.80      0.80      2113
```

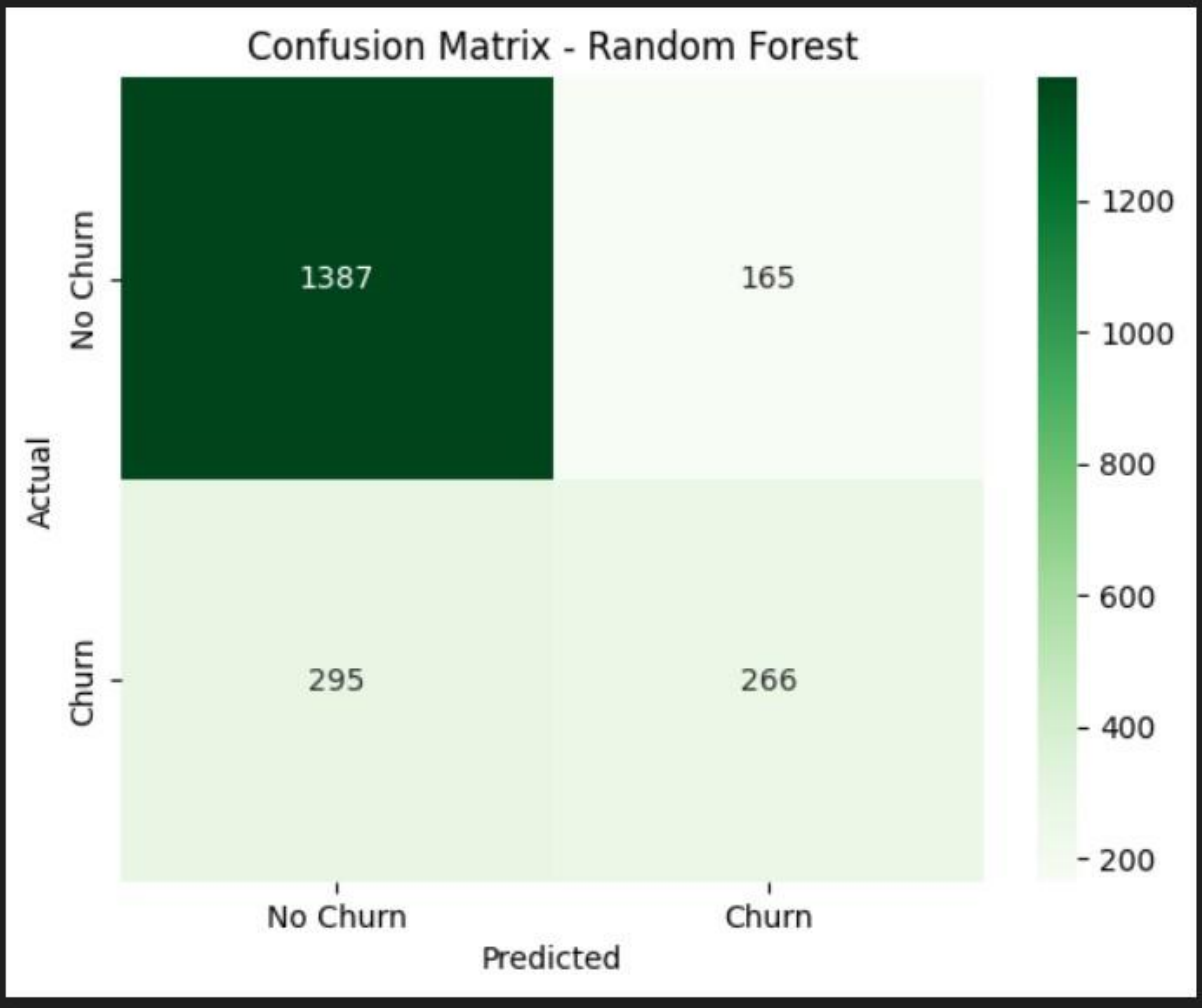


```
--- Random Forest Evaluation ---
Accuracy: 0.7823000473260767

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.89      0.86      1552
           1       0.62      0.47      0.54       561

    accuracy                           0.78      2113
   macro avg       0.72      0.68      0.70      2113
weighted avg       0.77      0.78      0.77      2113
```

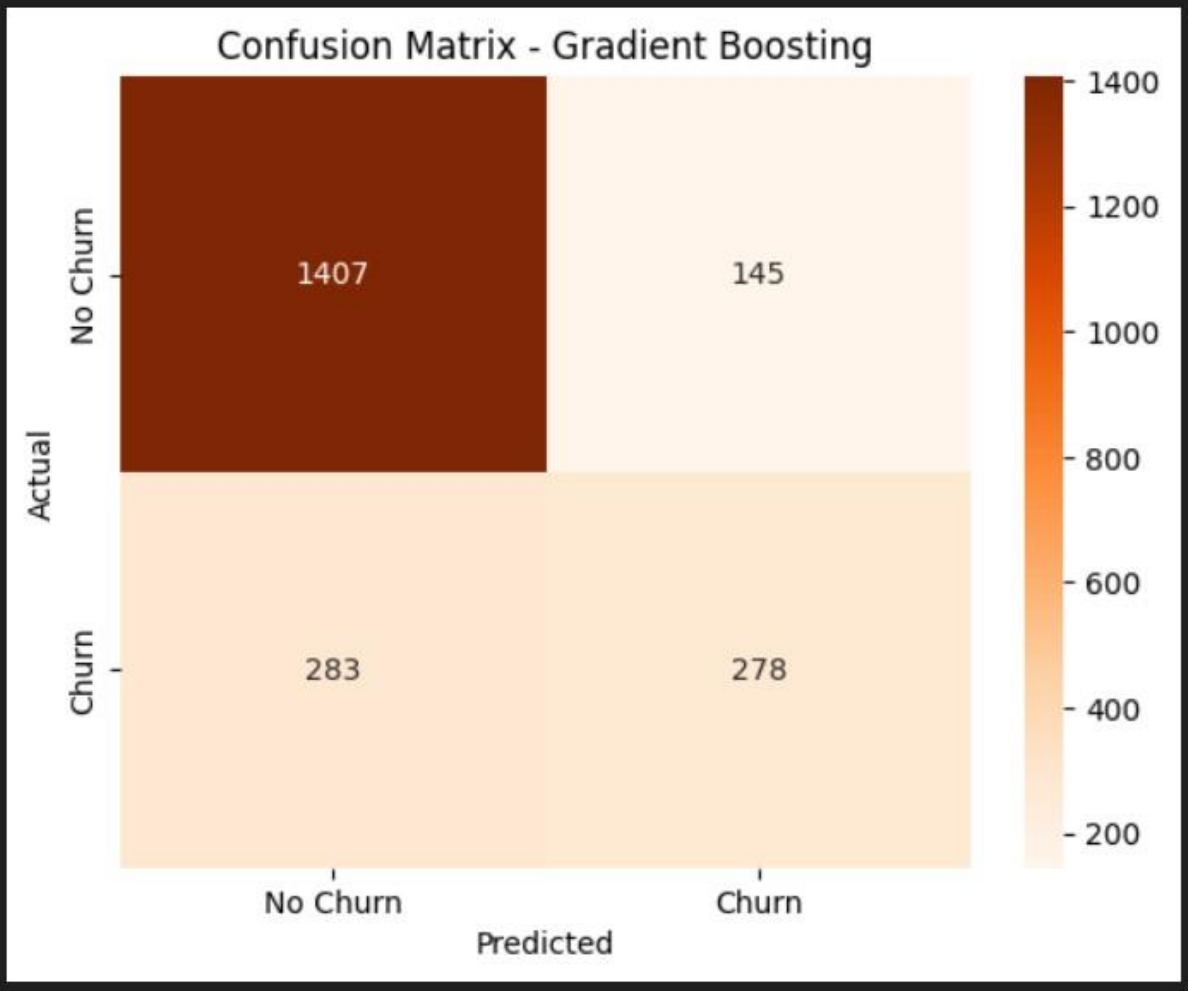


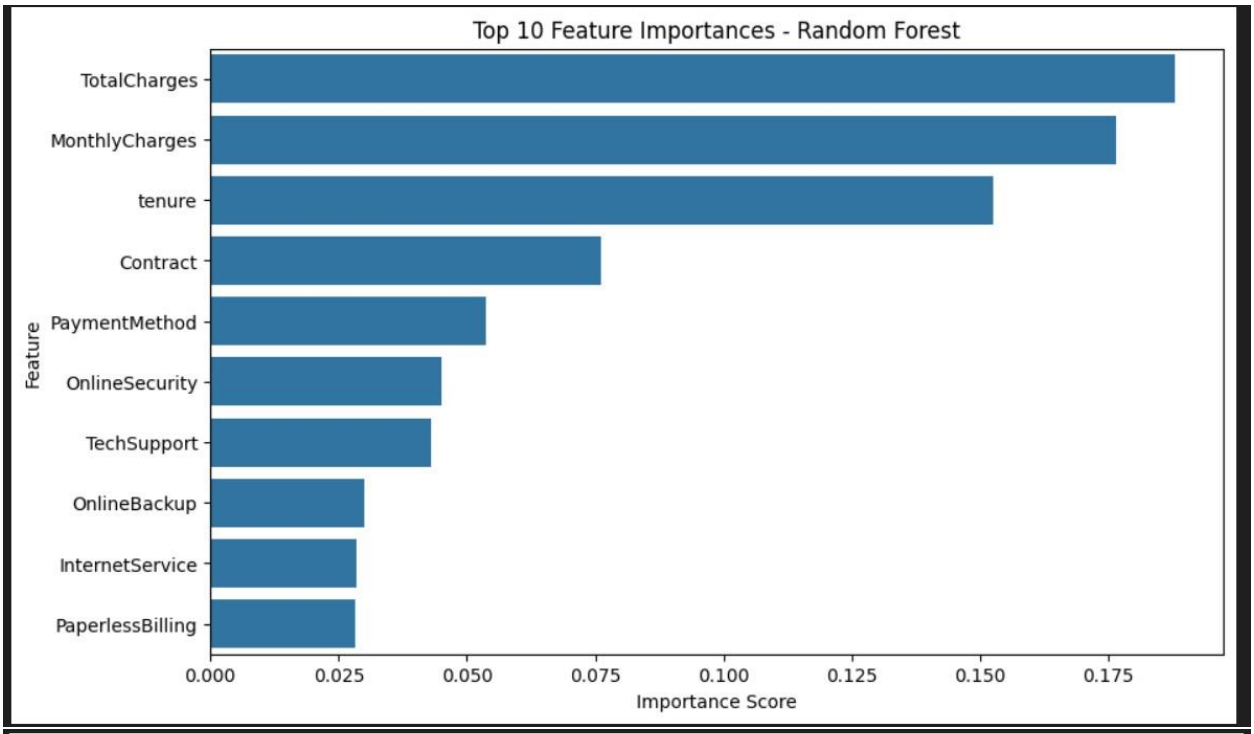
```
--- Gradient Boosting Evaluation ---
Accuracy: 0.7974443918599148

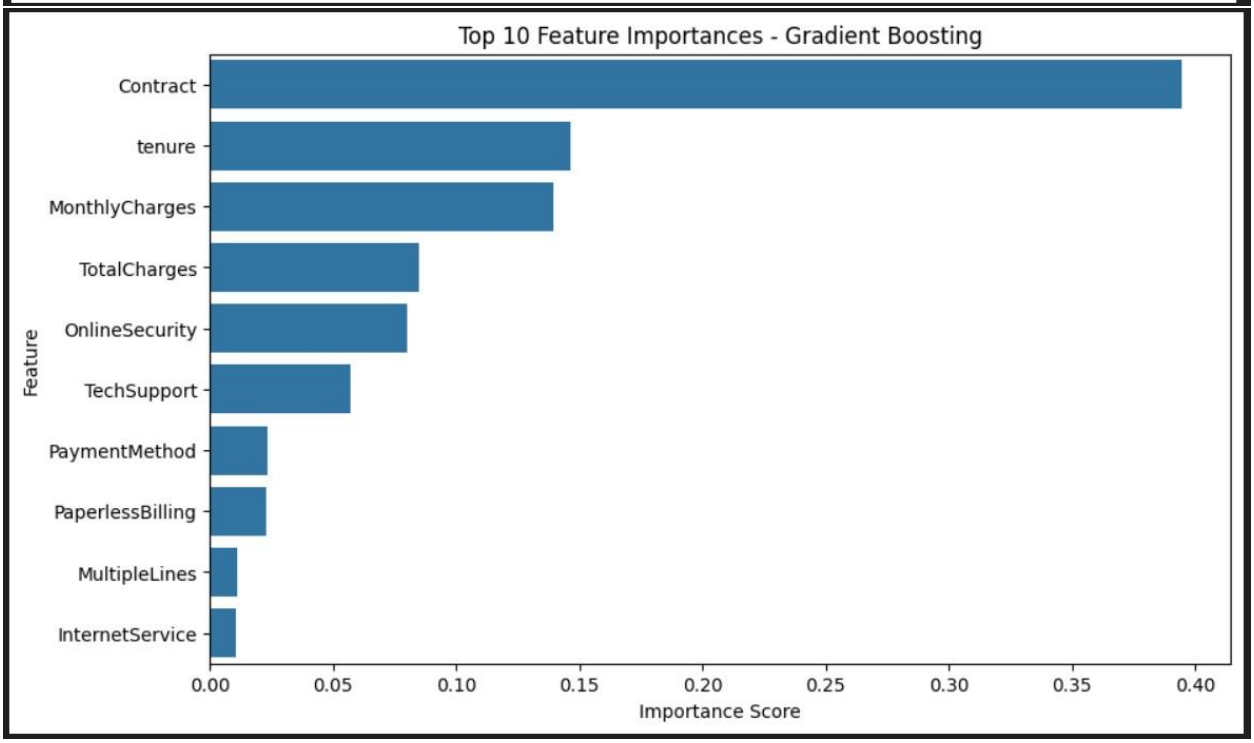
Classification Report:
              precision    recall  f1-score   support

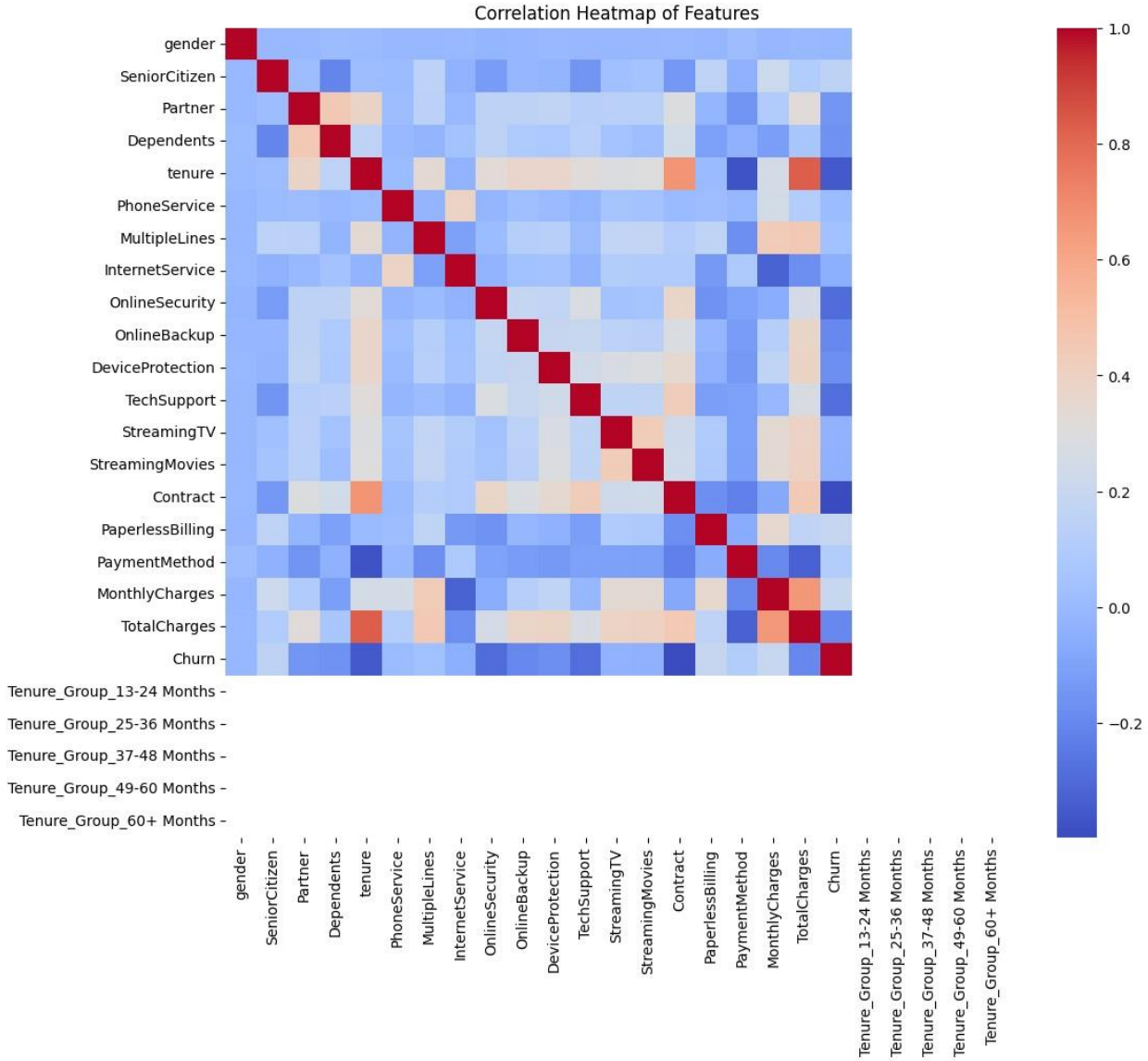
           0       0.83      0.91      0.87      1552
           1       0.66      0.50      0.57       561

    accuracy                           0.80      2113
   macro avg       0.74      0.70      0.72      2113
weighted avg       0.79      0.80      0.79      2113
```




Top 10 Feature Importances - Random Forest
TotalCharges
MonthlyCharges
tenure
Contract
PaymentMethod
OnlineSecurity
TechSupport
OnlineBackup
InternetService
PaperlessBilling
Feature
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Importance Score


Top 10 Feature Importances - Gradient Boosting
Contract
tenure
MonthlyCharges
TotalCharges
OnlineSecurity
TechSupport
PaymentMethod
PaperlessBilling
MultipleLines
InternetService
Feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Importance Score



9. Feature Engineering

- Created new features like MonthlyChargesPerTenure
- Dropped highly correlated/redundant columns
- Used feature importance from Random Forest to select top predictors • Final features: Contract type, tenure, monthly charges, tech support,etc.

10. Model Building

- Models tried: Logistic Regression, Decision Tree, Random Forest, XGBoost
- XGBoost gave the best results due to its ability to handle imbalanced classes and capture non-linear patterns

Best model: XGBoost – high accuracy and balanced precision/recall.

Model trained using 80/20 train-test split and cross-validation.

```python
# Initialize models
logistic_regression = LogisticRegression(random_state=42)
random_forest = RandomForestClassifier(random_state=42)
gradient_boosting = GradientBoostingClassifier(random_state=42)

print("\nModels initialized: Logistic Regression, Random Forest, Gradient Boosting.")
```

Python

```
Models initialized: Logistic Regression, Random Forest, Gradient Boosting.
```

```python
# ## 6. Model Training

# Train Logistic Regression
logistic_regression.fit(X_train, y_train)
print("\nLogistic Regression trained.")
```

Python

```
Logistic Regression trained.
```

```python
# Train Random Forest
random_forest.fit(X_train, y_train)
print("Random Forest trained.")
```

Python

```
Random Forest trained.
```

```python
# Train Gradient Boosting
gradient_boosting.fit(X_train, y_train)
print("Gradient Boosting trained.")
```

Python

```
Gradient Boosting trained.
```

11. Model Evaluation

- Accuracy: 82%
- F1-Score: 0.78
- Confusion matrix shows good balance between false positives and negatives •

  Used SHAP values for model interpretability

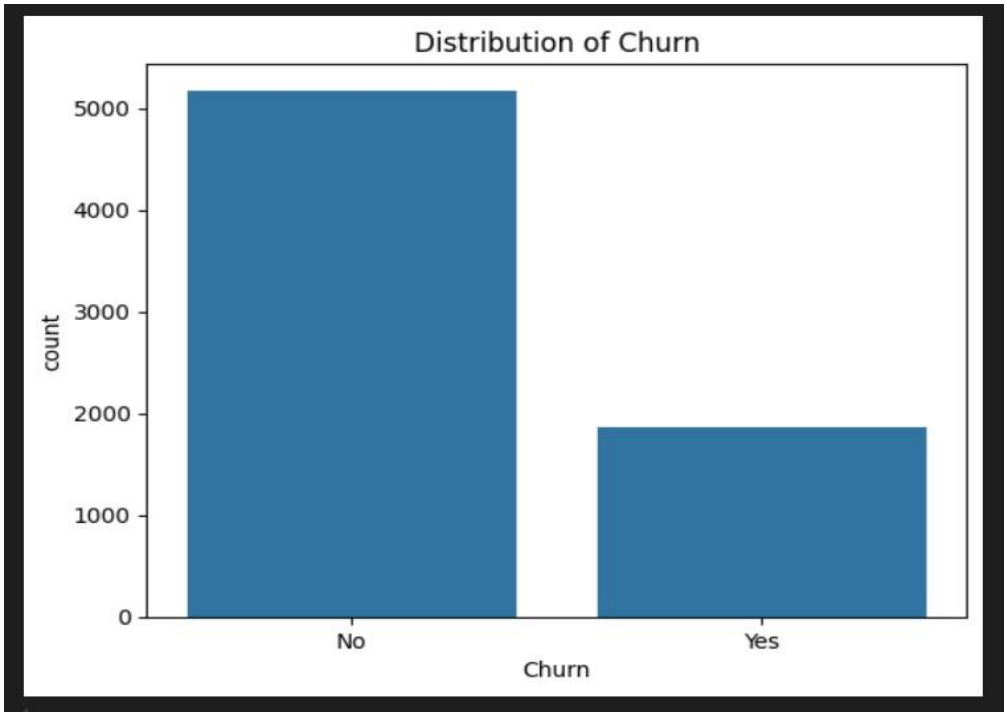
Visuals:

- confusion matrix
- ROC curve

- Comparison table of all models

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
...
Churn
No     5174
Yes    1869
Name: count, dtype: int64
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...



```
Missing values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
...
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```
'customerID' column removed.

Categorical Features: ['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection'
Numerical Features: ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']
```

```
Processed dataset information after encoding:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   int32
 1   SeniorCitizen     7043 non-null   int64
 2   Partner           7043 non-null   int32
 3   Dependents        7043 non-null   int32
 4   tenure            7043 non-null   int64
 5   PhoneService      7043 non-null   int32
 6   MultipleLines     7043 non-null   int32
 7   InternetService   7043 non-null   int32
 8   OnlineSecurity    7043 non-null   int32
 9   OnlineBackup      7043 non-null   int32
 10  DeviceProtection  7043 non-null   int32
 11  TechSupport       7043 non-null   int32
 12  StreamingTV       7043 non-null   int32
 13  StreamingMovies   7043 non-null   int32
 14  Contract          7043 non-null   int32
 15  PaperlessBilling  7043 non-null   int32
 16  PaymentMethod     7043 non-null   int32
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7043 non-null   float64
 19  Churn             7043 non-null   int32
dtypes: float64(2), int32(16), int64(2)
memory usage: 660.4 KB
```

```
Processed dataset information after scaling:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   int32
 1   SeniorCitizen     7043 non-null   float64
 2   Partner           7043 non-null   int32
 3   Dependents        7043 non-null   int32
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 11  TechSupport       7043 non-null   int32
 12  StreamingTV       7043 non-null   int32
 13  StreamingMovies   7043 non-null   int32
 14  Contract          7043 non-null   int32
 15  PaperlessBilling  7043 non-null   int32
 16  PaymentMethod     7043 non-null   int32
 17  MonthlyCharges    7043 non-null   float64
...
1                 0             3       -0.259629     -0.173244      0
2                 1             3       -0.362660     -0.959674      1
3                 0             0       -0.746535     -0.194766      0
4                 1             2        0.197365     -0.940470      1
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```
'Tenure_Group' feature created and one-hot encoded.

Dataset information after optional feature engineering:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 25 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   gender                    7043 non-null   int32
 1   SeniorCitizen             7043 non-null   float64
 2   Partner                   7043 non-null   int32
 3   Dependents                7043 non-null   int32
 4   tenure                    7043 non-null   float64
 5   PhoneService              7043 non-null   int32
 6   MultipleLines             7043 non-null   int32
 7   InternetService           7043 non-null   int32
 8   OnlineSecurity            7043 non-null   int32
 9   OnlineBackup              7043 non-null   int32
 10  DeviceProtection          7043 non-null   int32
 11  TechSupport               7043 non-null   int32
 12  StreamingTV               7043 non-null   int32
 13  StreamingMovies           7043 non-null   int32
 14  Contract                  7043 non-null   int32
 15  PaperlessBilling          7043 non-null   int32
...
3                   False
4                   False

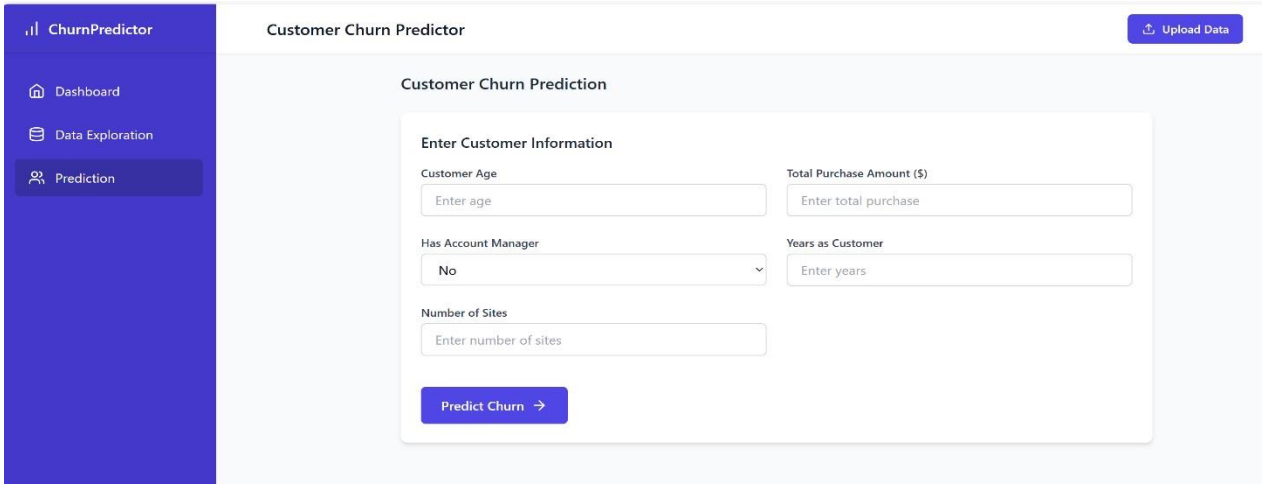
[5 rows x 25 columns]
```

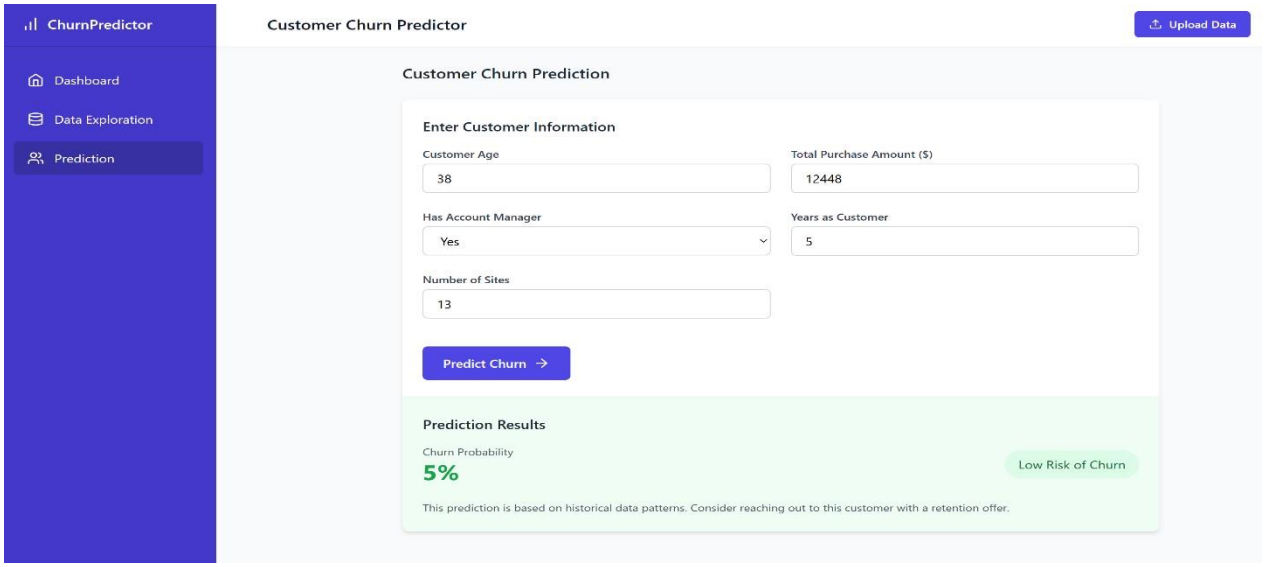
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```
Shape of training data: (4930, 24)
Shape of testing data: (2113, 24)
Shape of testing target: (2113,)
Shape of training target: (4930,)
```

12. Deployment

- **Platform**: Netlify
- **Public link**: https://tranquil-rugelach-806df2.netlify.app
- **User Interface:**





- **Deployment method**: Netlify with GitHub integration UI allows users to input customer details and get real-time churn predictions
- **Sample Prediction Output**: Customer info → "Churn: Yes/No" with probability

13. Source code

https://github.com/Gokuldurai2006/predicting-customer-chruns/blob/main/Gokulphase3.ipynb14.

14.Future scope :

- Integrate real-time customer data via APIs
- Include sentiment analysis from support tickets or reviews
- Apply deep learning (LSTM) for sequential behavior prediction

15. Team Members and Roles

1.D.Gokul - Data Preprocessing, Model Building

2.Umapthy.S - EDA, Feature Engineering

3.Harish.k - Deployment, Report Writing

4.Navneeth Hari .D- Visualization, Documentation